

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Data Structures Practical

Practical - V

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Binary Search Tree: Write a program to:

- a. Create a binary search tree.

Code:

```
class TreeNode:
    def __init__(self, data):
        self.data = data
        self.left = None
        self.right = None
```

```
root = TreeNode('R')
nodeA = TreeNode('A')
nodeB = TreeNode('B')
nodeC = TreeNode('C')
nodeD = TreeNode('D')
nodeE = TreeNode('E')
nodeF = TreeNode('F')
nodeG = TreeNode('G')
```

```
root.left= nodeA
root.right= nodeB
```

```
nodeA.left= nodeC
nodeA.right= nodeD
```

```
nodeB.left = nodeE
nodeB.right = nodeF
```

```
nodeF.left = nodeG
```

```

print("root.right.left.data:", root.right.left.data)
print("root.left.left.data:", root.left.left.data)
print("root.right.right.left.data:", root.right.right.left.data)

```

Output:

```

DS/ptacs.py
root.right.left.data: E
root.left.left.data: C
root.right.right.left.data: G
>

```

- b. Insert nodes into a binary search tree.

Code:

```

#Insert
nodeH= TreeNode('H')
nodeF.left= nodeG
nodeF.right= nodeH

print(" ", root.right.right.right.data)

```

Output:

```

root.right.left.data: E
H

```

- c. Search for a node in a binary search tree.

Code:

```

#DFS(depth for search)
def search(root, key):
    if root is None:
        return False

    print("Checking:", root.data)

    if root.data == key:
        return True

    return search(root.left, key) or search(root.right, key)

```

```

key = input("Enter element to search: ")

```

```
# Search
if search(root, key):
    print("Element Found")
else:
    print("Element Not Found")
```

Output:

```
root.right.left.data: E
Enter element to search: G
Checking: R
Checking: A
Checking: C
Checking: D
Checking: B
Checking: E
Checking: F
Checking: G
Element Found
```

```
root.right.left.data: E
Enter element to search: M
Checking: R
Checking: A
Checking: C
Checking: D
Checking: B
Checking: E
Checking: F
Checking: G
Element Not Found
```

